ACTIVITY SHEET – 5

Programs Using Constructors and Method Overloading

1. OBJECTIVES

By completing this activity, students will be able to:

• Understand the concept of constructors in Java
• Differentiate between default and parameterized constructors
• Apply constructor overloading
• Implement method overloading
• Improve object initialization and program design skills

2. READING REQUIREMENTS

Oracle Java Tutorials – Constructors:
https://docs.oracle.com/javase/tutorial/java/javaOO/constructors.html

Oracle Java Tutorials – Method Overloading:
https://docs.oracle.com/javase/tutorial/java/javaOO/methods.html

3. ILLUSTRATIVE EXAMPLES

Example 1: Constructor and Constructor Overloading

Code:

```
class Student {
    String name;
    int rollNo;

    Student() {
        name = "Unknown";
        rollNo = 0;
    }

    Student(String n, int r) {
        name = n;
        rollNo = r;
    }

    void display() {
```

```
        System.out.println(name + " " + rollNo);
    }
}

public class TestStudent {
    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student("Anita", 101);
        s1.display();
        s2.display();
    }
}
```

Example 2: Method Overloading

Code:

```
class Calculator {
    int add(int a, int b) {
        return a + b;
    }

    double add(double a, double b) {
        return a + b;
    }
}

public class TestCalc {
    public static void main(String[] args) {
        Calculator c = new Calculator();
        System.out.println(c.add(5, 3));
        System.out.println(c.add(2.5, 3.5));
    }
}
```

4. PROGRAMMING / TASK QUESTIONS

A. EASY LEVEL

1. Create a class Book with title and price. Initialize values using a constructor and display details.

Hint: Use a parameterized constructor.

JAVA CODE:

```
Main_Book.java ×   Main_Student_Details.class   Main_Student_Deta
Main_Book.java > ...
import java.util.Scanner;
class Book {
    String title;
    double price;
    Book(String title, double price) {
        this.title = title;
        this.price = price;
    }
    void displayDetails() {
        System.out.println(x: "\n--- Book Details ---");
        System.out.println("Title : " + title);
        System.out.println("Price : " + price);
    }
}
public class Main_Book {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter book title: ");
        String title = sc.nextLine();
        System.out.print(s: "Enter book price: ");
        double price = sc.nextDouble();
        Book b1 = new Book(title, price);
        b1.displayDetails();
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Book.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Book.java
Enter book title: JAVA PROGRAMMING
Enter book price: 40000

--- Book Details ---
Title : JAVA PROGRAMMING
Price : 40000.0
```

2. Create a class Rectangle with length and breadth. Use a constructor to initialize values and calculate area.
Hint: Define constructor and area() method.

JAVA CODE:

Main_Book.java | Main_Rectangle_Area.java | Main_Student_Details.class

Main_Rectangle_Area.java > ...

```java
import java.util.Scanner;
class Rectangle {
    int length;
    int breadth;
    Rectangle(int length, int breadth) {
        this.length = length;
        this.breadth = breadth;
    }
    int area() {
        return length * breadth;
    }
}
public class Main_Rectangle_Area {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter length: ");
        int length = sc.nextInt();
        System.out.print(s: "Enter breadth: ");
        int breadth = sc.nextInt();
        Rectangle r1 = new Rectangle(length, breadth);
        System.out.println("\nArea of Rectangle = " + r1.area());
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Rectangle_Area.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Rectangle_Area.java
Enter length: 30
Enter breadth: 20

Area of Rectangle = 600
```

3. Create a class Student with default constructor that assigns default values and displays them.
Hint:Use no-argument constructor.

JAVA CODE:

Main_Book.java | Main_Student_Details.class | Main_Student_Details.java

Main_Student_Details.java > ...

```java
class Student {
    String name;
    int rollNo;
    double cgpa;
    Student() {
        name = "Unknown";
        rollNo = 0;
        cgpa = 0.0;
    }
    void display() {
        System.out.println("--- Student Details (Default Values) ---");
        System.out.println("Name   : " + name);
        System.out.println("RollNo : " + rollNo);
        System.out.println("CGPA   : " + cgpa);
    }
}
public class Main_Student_Details {
    Run | Debug
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.display();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>Javac Main_Student_Details.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Student_Details.java
--- Student Details (Default Values) ---
Name   : Unknown
RollNo : 0
CGPA   : 0.0
```

4. A company maintains employee records. Some employees are created with only name, while others are created with name and salary. Design a program to handle both cases.
Hint: Constructor overloading is applicable.

JAVA CODE:

Main_Book.java | Main_Rectangle_Area.java | Main_Employee_Records.java

Main_Employee_Records.java > ...

```java
class Employee {
    String name;
    double salary;
    Employee(String name) {
        this.name = name;
        this.salary = 0;
    }
    Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }
    void displayDetails() {
        System.out.println("Name   : " + name);
        System.out.println("Salary : " + salary);
        System.out.println();
    }
}
public class Main_Employee_Records {
    Run | Debug
    public static void main(String[] args) {
        Employee e1 = new Employee(name: "Ravi");
        Employee e2 = new Employee(name: "Anita", salary: 45000);
        System.out.println(x: "--- Employee 1 Details ---");
        e1.displayDetails();
        System.out.println(x: "--- Employee 2 Details ---");
        e2.displayDetails();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Employee_Records.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Employee_Records.java
--- Employee 1 Details ---
Name   : Ravi
Salary : 0.0

--- Employee 2 Details ---
Name   : Anita
Salary : 45000.0
```

5. Design a program where a calculator performs addition for integers and floating-point numbers using same method name.
Hint: Method overloading is applicable.

JAVA CODE:

Main_Book.java | Main_Rectangle_Area.java | Main_Employee_Records.java | Main_Calcula

Main_Calculator_Addition.java > ...

```java
import java.util.Scanner;
class Calculator {
    int add(int a, int b) {
        return a + b;
    }
    double add(double a, double b) {
        return a + b;
    }
}
public class Main_Calculator_Addition {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Calculator c = new Calculator();
        System.out.print(s: "Enter two integers: ");
        int int1 = sc.nextInt();
        int int2 = sc.nextInt();
        int intResult = c.add(int1, int2);
        System.out.println("Integer Addition Result : " + intResult);
        System.out.print(s: "Enter two decimal numbers: ");
        double d1 = sc.nextDouble();
        double d2 = sc.nextDouble();
        double doubleResult = c.add(d1, d2);
        System.out.println("Floating-point Addition Result : " + doubleResult);
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Calculator_Addition.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Calculator_Addition.java
Enter two integers:
2
3
Integer Addition Result : 5
Enter two decimal numbers:
3.2
4.5
Floating-point Addition Result : 7.7
```

6. A mobile phone object can be created with only brand name or with brand name and price. Design a Java program to model this situation.
Hint: Constructor overloading is applicable.

JAVA CODE:

Main_Book.java | Main_Rectangle_Area.java | Main_Employee_Records.java

Main_Mobile_Details.java > Main_Mobile_Details > main(String[])

```java
import java.util.Scanner;
class Mobile {
    String brand;
    double price;
    Mobile(String brand) {
        this.brand = brand;
        this.price = 0;
    }
    Mobile(String brand, double price) {
        this.brand = brand;
        this.price = price;
    }
    void displayDetails() {
        System.out.println("Brand : " + brand);
        System.out.println("Price : " + price);
        System.out.println();
    }
}
public class Main_Mobile_Details {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter brand name: ");
        String brand = sc.nextLine();
        System.out.print(s: "Do you want to enter price? (yes/no): ");
        String choice = sc.nextLine();
        Mobile m;
```

```java
        if (choice.equalsIgnoreCase(anotherString: "yes")) {
            System.out.print(s: "Enter price: ");
            double price = sc.nextDouble();
            m = new Mobile(brand, price);
        } else {
            m = new Mobile(brand);
        }
        System.out.println(x: "\n--- Mobile Details ---");
        m.displayDetails();
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Mobile_Details.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Mobile_Details.java
Enter brand name: Samsung
Do you want to enter price? (yes/no): yes
Enter price: 50000

--- Mobile Details ---
Brand : Samsung
Price : 50000.0


C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Mobile_Details.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Mobile_Details.java
Enter brand name: samsung
Do you want to enter price? (yes/no): no

--- Mobile Details ---
Brand : samsung
Price : 0.0
```

C. HARD LEVEL (Design-Oriented)

7. A banking application allows creating an account with account number only, or with account number and initial balance. Design a Java program for this system.

Hint: Constructor overloading is applicable.

JAVA CODE:

Main_Book.java | Main_Rectangle_Area.java | Main_Employee_Records.java

Main_Bank_Account.java > ...

```java
import java.util.Scanner;
class BankAccount {
    int accountNumber;
    double balance;
    BankAccount(int accountNumber) {
        this.accountNumber = accountNumber;
        this.balance = 0;
    }
    BankAccount(int accountNumber, double balance) {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }
    void displayDetails() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Balance        : " + balance);
        System.out.println();
    }
}
```

```java
}
public class Main_Bank_Account {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter account number: ");
        int accNo1 = sc.nextInt();
        BankAccount a1 = new BankAccount(accNo1);
        System.out.print(s: "Enter account number: ");
        int accNo2 = sc.nextInt();
        System.out.print(s: "Enter initial balance: ");
        double bal = sc.nextDouble();
        BankAccount a2 = new BankAccount(accNo2, bal);
        System.out.println(x: "\n--- Account 1 Details ---");
        a1.displayDetails();
        System.out.println(x: "--- Account 2 Details ---");
        a2.displayDetails();
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Bank_Account.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Bank_Account.java
Enter account number: 12345
Enter account number: 45678
Enter initial balance: 9000

--- Account 1 Details ---
Account Number : 12345
Balance        : 0.0

--- Account 2 Details ---
Account Number : 45678
Balance        : 9000.0
```

8. Design a program to calculate area of shapes where area() method works for circle and rectangle using same method name.
Hint: Method overloading is applicable.

JAVA CODE:

Main_Area_Shapes.java > ...

```java
import java.util.Scanner;
class ShapeArea {
    double area(double radius) {
        return 3.14 * radius * radius;
    }
    double area(double length, double breadth) {
        return length * breadth;
    }
}
public class Main_Area_Shapes {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ShapeArea s = new ShapeArea();
        System.out.print(s: "Enter radius of circle: ");
        double r = sc.nextDouble();
        System.out.println("Area of Circle = " + s.area(r));
        System.out.print(s: "Enter length of rectangle: ");
        double l = sc.nextDouble();
        System.out.print(s: "Enter breadth of rectangle: ");
        double b = sc.nextDouble();
        System.out.println("Area of Rectangle = " + s.area(l, b));

        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Area_Shapes.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Area_Shapes.java
Enter radius of circle: 50
Area of Circle = 7850.0
Enter length of rectangle: 90
Enter breadth of rectangle: 80
Area of Rectangle = 7200.0
```

9. Create a class Temperature that converts temperature from Celsius to Fahrenheit and Kelvin using overloaded methods.
Hint: Use method overloading with different parameter lists.

JAVA CODE:

```
_Calculator_Addition.java   Main_Mobile_Details.java   Main_Bank_Account.java   Main_Area
Main_Temperature_Conversion.java > ...
import java.util.Scanner;
class Temperature {
    double convert(double celsius) {
        return (celsius * 9 / 5) + 32;
    }
    double convert(double celsius, int k) {
        return celsius + 273.15;
    }
}
public class Main_Temperature_Conversion {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Temperature t = new Temperature();
        System.out.print(s: "Enter temperature in Celsius: ");
        double c = sc.nextDouble();
        System.out.println("Temperature in Fahrenheit = " + t.convert(c));
        System.out.println("Temperature in Kelvin = " + t.convert(c, k: 1));
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>javac Main_Temperature_Conversion.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\Activity_sheet_5.java>java Main_Temperature_Conversion.java
Enter temperature in Celsius: 300
Temperature in Fahrenheit = 572.0
Temperature in Kelvin = 573.15
```

5. ASSIGNMENT EVALUATION

0 : Not Done
1 : Incomplete
2 : Late Completion
3 : Needs Improvement
4 : Complete
5 : Well Done

6. VIVA QUESTIONS

1. What is a constructor?

2. What is the difference between default and parameterized constructor?

3. What is constructor overloading?

4. What is method overloading?

5. Can constructors have return type?

6. When is a constructor called?

7. What is the difference between constructor and method?